I Encuentro virtual de semilleros en Inteligencia Artificial
Universidad Santo Tomás, Tunja 🇨🇴

Fewer bits, same performance? An introduction to quantization



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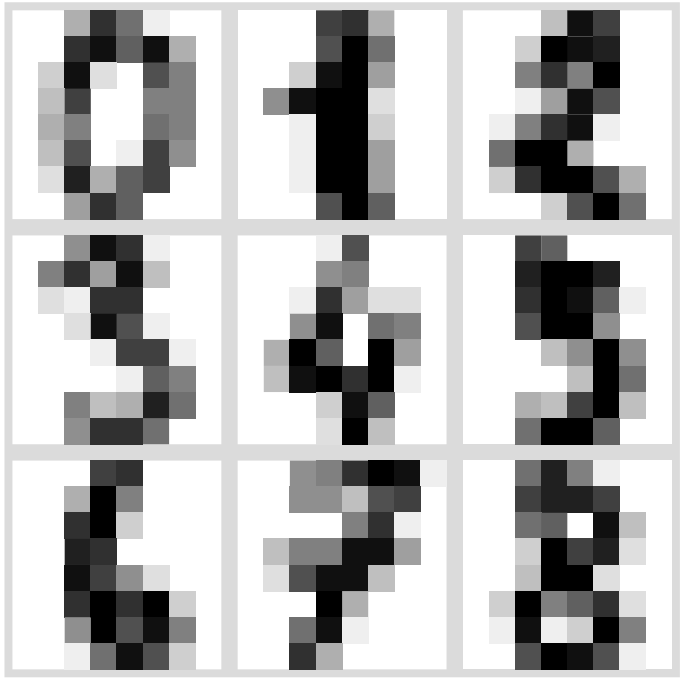
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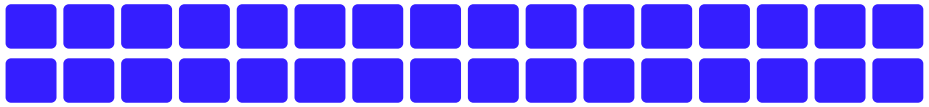
Research Question

Does an **int8** neural network, running on an ESP32,
keep the performance of its **float32** version?

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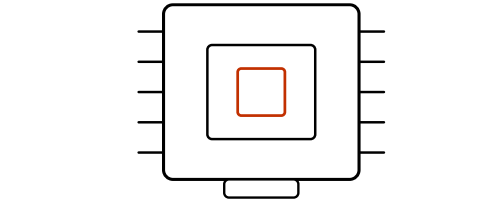
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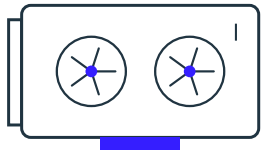
float32 32 bits · 4 bytes



Quantization →

int8 8 bits · 1 byte



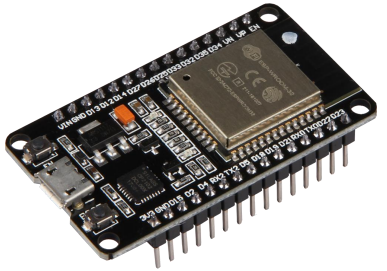


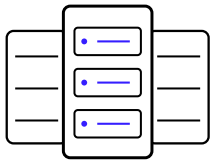
GPU (local)

Edge device (microcontroller)



Cloud





AI Factory

Image: Joy-IT, *NodeMCU ESP32* (joy-it.net)

Note: float32 and int8 were chosen for simplicity and according to the objectives of this presentation. Other numeric formats used in practice, such as bf16, fp8 or int4, are out of scope.

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02

Scope and Objectives

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Scope and Objectives

Scope

Classification of 8×8 digits with a small dense network, quantized to int8 after training and deployed on an ESP32.

General objective

Identify the **technical trade-offs** involved in bringing a neural network to a microcontroller, using **int8 quantization**.

Specific objectives

1. Identify the technical trade-offs of quantizing and deploying on a microcontroller.
2. Compare accuracy, size and latency between float32 and int8.
3. Validate the quantized model on the ESP32.

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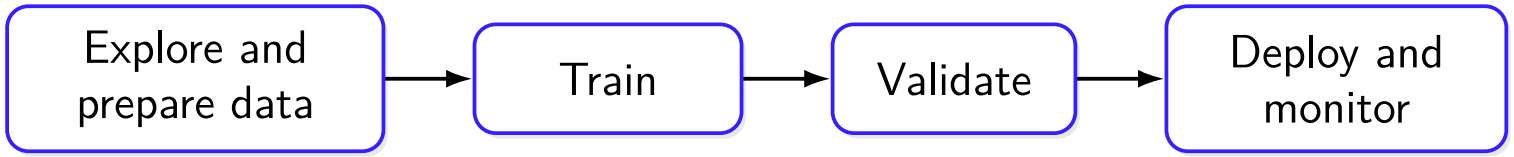
03

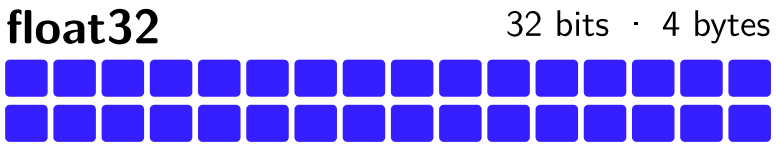
From float32 to int8

Answering the research question

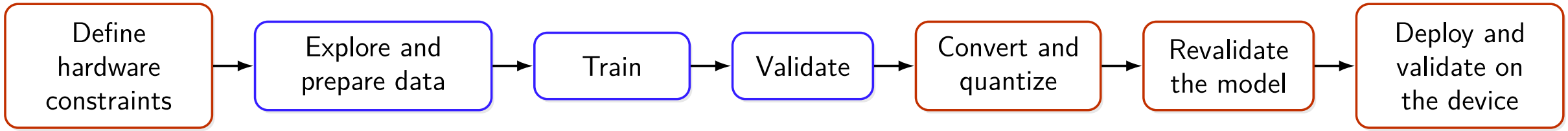
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Standard workflow

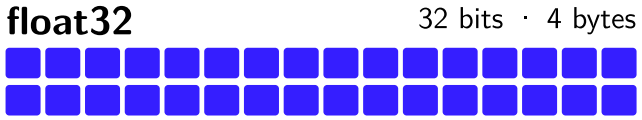




Microcontroller workflow



RAM: ~324 KB free
Flash: 4 MB
Compute: 2 cores 240 MHz



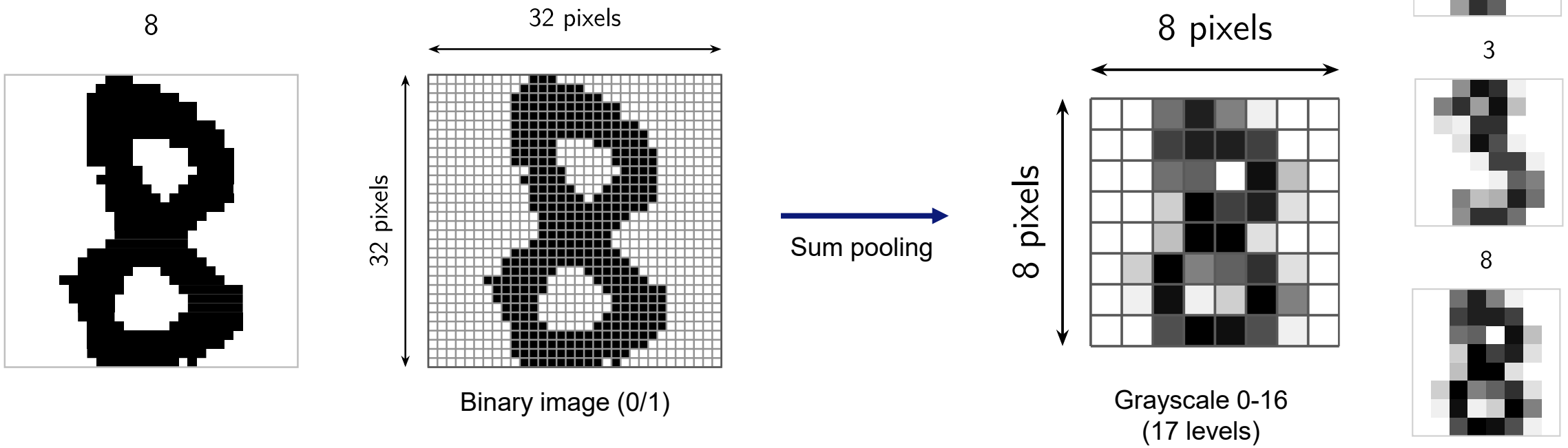


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Dataset

Optical Recognition of Handwritten Digits

- **Source:** Machine Learning Repository, UCI*
- **Authors:** Ethem Alpaydin and Cenk Kaynak (1998)
- **Number of images:** 1797
- **Classes:** 10 digits (0-9) [~180 per class]



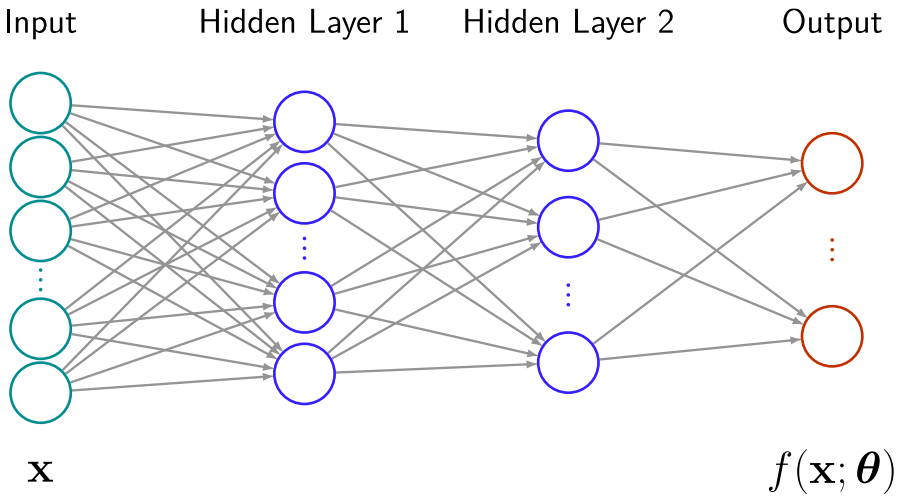
(*) University of California, Irvine

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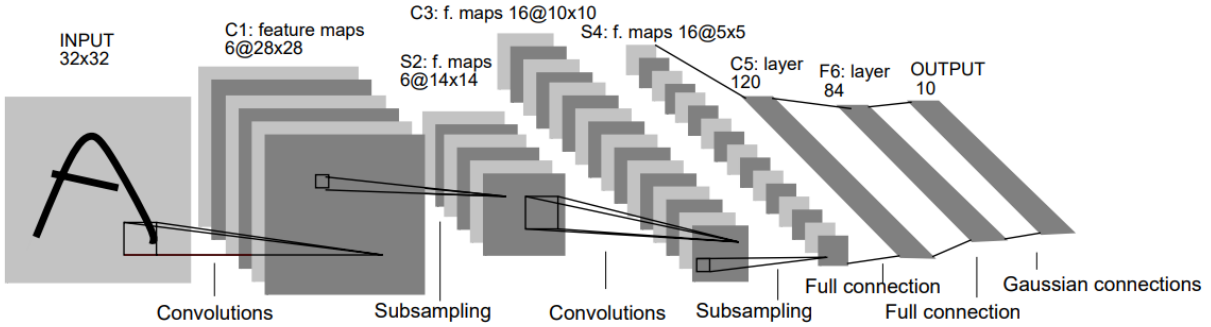
Neural Networks

Feedforward neural networks, also called **multilayer perceptrons (MLPs)**, are the quintessential deep learning models. Their goal is to approximate some function f^*: they define a mapping $\boldsymbol{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$ and learn the parameters $\boldsymbol{\theta}$ that result in the best function approximation.

— I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016, ch. 6 (adapted).



Feedforward Neural Network



LeNet-5 architecture, a **Convolutional Neural Network** for digit recognition.
***Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner**, "Gradient-based learning applied to document recognition,"* Proceedings of the IEEE, *vol. 86*, no. 11, *pp. 2278–2324*, *Nov. 1998*.
[Link to the paper]

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Hardware

ESP32

Specification	Value
Chip	ESP32-D0WD-V3 (revision 3.0)
Processor	Xtensa LX6 · 2 cores at 240 MHz
Floating point	Single precision (float32)
RAM (SRAM)	**520 KB total · ~324 KB free ****
Flash	**4 MB**
PSRAM	Not included
Connectivity	Wi-Fi 802.11 b/g/n · Bluetooth 4.2
PC connection	USB (chip CP210x)

Source: Espressif, ESP32 datasheet.
(**) Own measurement in Arduino IDE (free RAM)

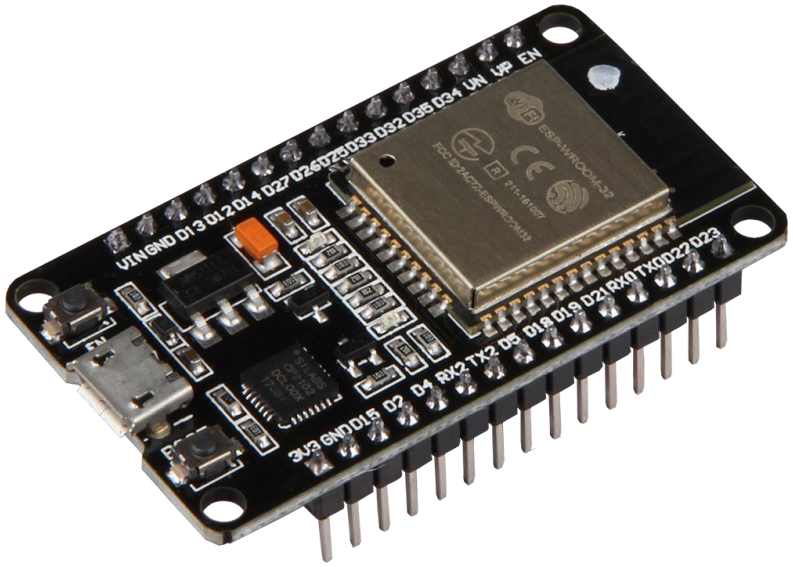


Image: Joy-IT, *NodeMCU ESP32* (joy-it.net)

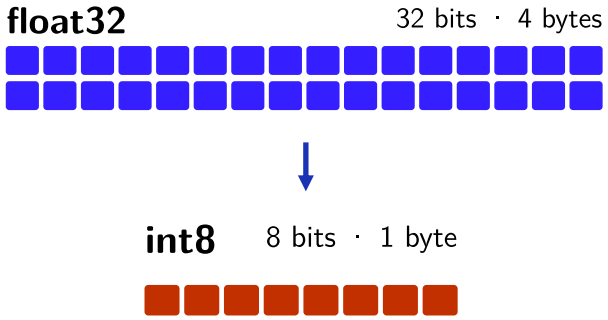
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Quantization

What It Is, What It Costs and Why

01 What it is

Representing numbers with fewer bits



$$q = \text{round}\left(\frac{x}{s}\right) + z$$

s: scale
z: zero point

02 What it costs

Accuracy ⇄ memory
Rounding introduces error

Range ⇄ resolution
An outlier degrades the scale

Calibration
Requires representative data

Revalidation
It is a different model: measure it again

03 Why int8

4×

less memory per weight

- **More free RAM** for the rest of the application
- **Optimized operations** for int8
- **Scales** to larger models

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From float32 to int8

Let's Experiment!

Notebook: Introduction to Quantization

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Questions?

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